1.3.15 HSF

Short for Heat Sink Fan unit, this refers to the aluminium block and fan unit placed over the CPU to aid in dissipation of heat. While most stock HSF units are made of aluminium and use a fan, fancier cooling solutions made of copper and using liquid coolants are also available.

Screenshot 17: Heatsink Fan unit.jpeg

1.3.16 Overclocking

Overclocking refers to the art of making a computer component perform at levels exceeding those set by the manufacturer. Components like the CPU, RAM, and graphics cards can be overclocked.

1.3.17 HyperThreading

This was a technology used by Intel to improve CPU utilisation by splitting a processing workload into parallel threads. Thanks to HyperThreading, the CPU appeared to the OS as multi-core. To make use of this feature, it was essential that the OS and motherboard supported it.

1.3.18 TDP

Thermal Design Profile refers to the average energy (expressed in watts) in the form of heat that the HSF has to dissipate from the CPU. This can be taken as a rule-of-thumb measure of the energy consumed by the CPU.

1.3.19 Transistor count

This refers to the number of transistors included in the CPU core. The transistor count is influenced not just by the complexity of the core, but also the size of the on-die cache. The transistor count of a Intel Core 2 Duo CPU with a 2 MB cache is 151.6 million.